Lesson plan · for the teacher

Distributions and histograms

Data science · Level 2 · Lesson 1

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

Learning goals

- Build a histogram with equal bins.
- Read symmetric and skewed shapes.
- Choose a middle that fits the shape.

Activity: A sticky-note histogram

1. Warm-up (5 min): Show sixty heights as a list and ask what pupils can tell from it. Not much.
2. Teach (8 min): Draw bins on the board, count a few heights into them, and sketch a hill and a skewed shape.
3. Build (15 min): Pupils measure hand spans and stick a note in the right bin. Then redraw with bins twice as wide.
4. Describe (7 min): Which width shows the shape best? Is it a hill or skewed? Which middle fits?
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- Why must the bins be equal?
- What did wider bins hide?
- Which school data might have two hills?

Exit ticket

- Sketch a skewed-right histogram and mark roughly where the mean and the median are.

Try it at school tomorrow
Measure hand spans, agree on bins together and build a histogram on the board with sticky notes. Is it a hill, or does it have a tail?

Practice questions · for you

Distributions and histograms

Data science · Level 2 · Lesson 1

1 **In a histogram of test marks, what does the height of each bar show?**

◯ The average mark ◯ The best mark ◯ How many pupils have marks in that bin

2 **In a histogram, the height of each bar is a count of values.**

◯ True ◯ False

3 **Count the heights into five bins of 5 cm, starting at 140 cm, and print the counts.**

```
heights = [142, 147, 151, 153, 156, 158, 161, 149, 152, 155]
counts = [0, 0, 0, 0, 0]
for h in heights:
    counts[0] = counts[0] + 1
print(counts)
```

Write your program:

4 **Match each dataset to its likely shape.**

Heights of 12-year-olds	• • • •	Two hills
Minutes to walk to school	• • • •	Symmetric hill
Heights of a class of 6-year-olds and their parents	• • • •	Skewed right

5 **In data that is skewed right, where is the mean?**

◯ Above the median ◯ Below the median ◯ Exactly on the median

✎ **Which data at your school do you think is skewed? Which middle have you been using for it?**

Check your answers

Distributions and histograms

Data science · Level 2 · Lesson 1

1 **How many pupils have marks in that bin**

Each bar counts the pupils whose marks fall in its bin.

2 **True**

Each bar shows how many values fall in its bin.

3

```
heights = [142, 147, 151, 153, 156, 158, 161, 149, 152, 155]
counts = [0, 0, 0, 0, 0]
for h in heights:
    b = (h - 140) // 5
    counts[b] = counts[b] + 1
print(counts)
```

(h - 140) // 5 turns each height into a bin number from 0 to 4.

4 **Heights of 12-year-olds — Symmetric hill**
Minutes to walk to school — Skewed right
Heights of a class of 6-year-olds and their parents — Two hills

Most heights cluster around a middle; journeys have a long tail; two very different groups make two hills.

5 **Above the median**

The long tail of large values pulls the mean up, above the median.